

Soln. SU24 sec34quiz1 Set A

$$\textcircled{1} \textcircled{a} \text{ Total} = 9 \times 10^8$$

$$X = 9 \times 10^8 \times \frac{15}{100} \\ = 1.35 \times 10^8$$

$$Y = 9 \times 10^8 \times \frac{15}{100} \\ = 1.35 \times 10^8$$

$$Z = 9 \times 10^8 \times \frac{40}{100} \\ = 3.6 \times 10^8$$

$$W = 9 \times 10^8 \times \frac{30}{100} \\ = 2.7 \times 10^8$$

Total time,

$$\text{AMD} = \frac{1.35 \times 10^8 \times 2 + 1.35 \times 10^8 \times 1 + 3.6 \times 10^8 \times 3 + 2.7 \times 10^8 \times 2}{2.5 \times 10^9 \text{ (CF)}}$$

(Sum of all IC x CPI)

$$= 0.81 \text{ sec}$$

$$\text{RYZ} = \frac{1.35 \times 10^8 \times 1 + 1.35 \times 10^8 \times 2 + 3.6 \times 10^8 \times 1 + 2.7 \times 10^8 \times 2}{3.5 \times 10^9}$$

$$= 0.37 \text{ sec}$$

$$\text{CPI (AMD)} = \frac{0.81 \times 2.5 \times 10^9}{9 \times 10^8} = 2.25 \quad (\text{Ans})$$

$$\text{CPI (RYZ)} = \frac{0.37 \times 3.5 \times 10^9}{9 \times 10^8} = 1.438 \quad (\text{Ans})$$

⑥ Clock Cycle (AMD)

$$= 1.35 \times 10^8 \times 2 \leftarrow X$$

$$+ 1.35 \times 10^8 \times 1 \leftarrow Y$$

$$+ 3.6 \times 10^8 \times 3 \leftarrow Z$$

$$+ 2.7 \times 10^8 \times 2 \leftarrow W$$

$$\begin{aligned} \text{Clock Cycle (Ryzen)} = & 1.35 \times 10^8 \times 1 \\ & + 1.35 \times 10^8 \times 2 \\ & + 3.6 \times 10^8 \times 1 \\ & + 2.7 \times 10^8 \times 2 \end{aligned}$$

⑦ Exec(AMD) = 0.81 sec  
Exec(Ryzen) = 0.37 sec

AMD SPEC ratio =  $\sqrt[1]{\frac{17800}{0.81}}$  or  $\sqrt[1]{\frac{0.81}{17800}}$

Ryzen SPEC ratio =  $\sqrt[1]{\frac{17800}{0.37}}$  or  $\sqrt[1]{\frac{0.37}{17800}}$

(Ans) → we will use 1 as we are only taking 1 value as parameter. This one has no significance

Due to this being a ratio, it won't matter whether we use  $\frac{\text{reference}}{\text{Execution}}$  or  $\frac{\text{Execution}}{\text{reference}}$

② diameter 32cm radius = 16cm

$$\text{die area} = \frac{\text{wafer area}}{\text{dies per wafer}} = \frac{\pi \times 16^2}{88}$$

$$= 9.1391 \text{ cm}^2$$

$$\text{Yield} = \frac{1}{\left(1 + (\text{defects per area} \times \text{Die area}/2)\right)^2}$$

$$= \frac{1}{\left(1 + (\text{defects per area} \times 9.1391/2)\right)^2}$$

$$= 0.6626$$

$$\text{Cost per die} = \frac{26}{88 \times 0.6626} = 0.4459$$

(Ans)

③ (a) Algorithm, Programming language, Compiler, ISA/Processor & memory system, I/O system.

(b) Datapath, Control, Cache memory

(c) Elapsed time: Total response time, including all aspects (I/O, OS overhead, idle time)

Execution time: only time spent processing given job

(d) (i) Reduce number of clock cycles

(ii) increase clock rate

(iii) trade off clock rate against cycle count

(e) Every 2 years, # of transistors double, size, computing power doubles.